Retrieval-Augmented Generation (RAG)

Why RAG? LLM Limitations, External Knowledge and Basic Architecture

Giving an LLM the Right Information at the Right Time

MODULE 2 · LECTURE 1 | DURATION · 60 MIN | SYLLABUS · 2.1 INTRODUCTION TO RAG

TOPICS COVERED

- Standalone LLM Recall
- Parametric Knowledge
- Knowledge Freshness
- Private Knowledge
- Hallucination
- Traceability
- External Knowledge
- RAG Concept
- Retriever & Generator
- RAG vs Prompting / FT / Search
- When to Use RAG
- Basic Workflow

WHERE ARE WE IN THE COURSE?

From Adapting the Model to Connecting It to Knowledge

MODULE 1

Fine-Tuning LLMs

How do we adapt model behavior?

MODULE 2

Retrieval-Augmented Generation

How do we provide external knowledge?

LECTURE 1 TODAY

Why RAG?

LECTURE 2

Complete RAG Pipeline

LECTURE 3

Retrieval Quality & Evaluation

LECTURE 4

Real-World Applications

LECTURE 5

Practical RAG Implementation

> "Module 1 was largely about adapting the model. Module 2 is about connecting the model to information outside its parameters."

Hook → LLM limitations → RAG idea → Architecture → RAG vs alternatives → When to use → Quiz + bridge

START WITH A REAL PROBLEM · UNIVERSITY ASSISTANT EXAMPLE

A Question the LLM Should Not Be Able to Answer

ASK AN LLM

"What is the last date for Semester VIII examination registration announced by our university yesterday?"

Now ask the students: "Is the model unintelligent?"

THE MODEL MAY…

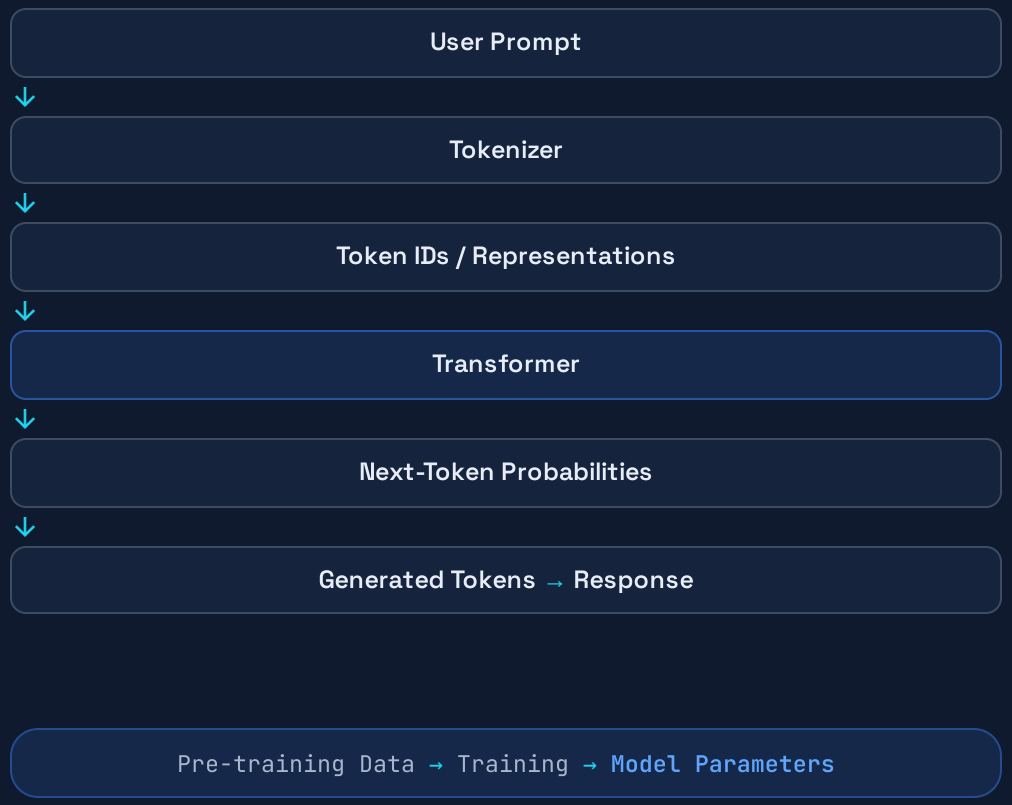
- not know the circular
- know an older date
- guess a date
- give a generic answer

No — the model is not unintelligent. The problem is: **the required information may not exist inside the model's learned parameters.**

RECALL FROM MODULE 1 · WHERE KNOWLEDGE LIVES

How a Basic LLM Answers — and What Parametric Knowledge Means

GENERATION PIPELINE



Pre-training Data → Training → Model Parameters

Parametric Knowledge

Information and patterns represented **implicitly in the model's learned parameters** — conceptually:

Training Data → Training → Billions of Parameters → Knowledge Patterns

A PRE-TRAINED LLM MAY KNOW

- ✓ general language
- ✓ common historical facts
- ✓ programming patterns
- ✓ common concepts
- ✓ broad world knowledge

Key idea: the model's parameters are **not a conventional searchable database** of sentences.

Important: a basic LLM does **not** automatically search your university database, latest PDF, company portal, or the Internet merely because you asked a question. External tools / retrieval must be integrated separately.

LIMITATION 1 / 5 THE FIVE LIMITATIONS

Knowledge Freshness

BLACKBOARD ITEM 1

LLM knowledge ≠ always current

SUPPOSE THE MODEL WAS TRAINED BEFORE A NEW EVENT

Ask: "Who won an event that happened yesterday?"

Yesterday's Event ✕ Model Parameters

The information may simply **not be available**.

Takeaway: static learned knowledge can become **stale**.

KNOWLEDGE THE MODEL MAY LACK

- today's stock information
- latest examination circular
- latest university policy
- newly released product manual
- today's support outage
- current regulation

LIMITATION 2 / 5 THE FIVE LIMITATIONS

Private / Organization-Specific Knowledge

IMAGINE THESE DOCUMENTS

- Company HR Policy.pdf
- Internal SOP
- Private Product Documentation
- University Examination Circular
- Hospital Internal Protocol

WERE THESE IN PRE-TRAINING DATA?

No.

Private documents are written **inside organizations** — they were never published for a crawler or a corpus to see.

General LLM ✕ **Automatic access to private organizational knowledge**

"How should a model know **our internal leave policy** if we never provide it?" — **Exactly.** It cannot.

Hallucination — and Why It Happens

WHAT IS HALLUCINATION?

A generated answer that **appears plausible** but contains **unsupported, incorrect, or fabricated** information.

EXAMPLE QUESTION

"What is the penalty mentioned in Circular CE/2026/147?"

If the model does not have the circular, it might still produce a **confident-looking answer**.

⚠ **Important:** fluent language does **not** guarantee factual correctness — one reason grounding is valuable.

WHY DOES IT HAPPEN? · THE LLM IS OPTIMIZED TO PRODUCE PROBABLE CONTINUATIONS



Likely text ≠ Guaranteed factual evidence

DO NOT TEACH

~~"RAG completely eliminates hallucinations."~~

CORRECT

RAG can **reduce unsupported generation** when retrieval and prompting are good, but it does **not guarantee perfect factuality**.

No Traceability — and Updating Knowledge by Retraining Is Expensive

LIMITATION 4 Traceability

Model says: **"The examination deadline is 17 October."** — Student asks: **"According to which circular?"**

A standalone answer may not provide a **verifiable source**. For many applications we want:

`Answer` + `Supporting Evidence` + `Source`

WHERE THIS MATTERS

legal systems | finance | research | university regulations | enterprise support

Key idea: production users often need **evidence**, not merely fluent answers.

LIMITATION 5 Updating by Retraining Is Expensive

Suppose a university changes its examination deadline:

`15 October` → `18 October`

Would you really, every time one circular changes:

`Collect Training Data` → `Fine-Tune` → `Validate` → `Redeploy`

Usually **unnecessary**.

Better: update the document / knowledge base, then **retrieve current information**. This is where RAG becomes powerful.

THE FIVE LIMITATIONS, SUMMARIZED

The Fundamental Problem — and the Pivot Question

STANDALONE LLM

- Knowledge may be **stale**
- Private knowledge may be **missing**
- Can **hallucinate**
- Source **traceability** may be weak
- Updating facts via retraining is **costly**

THE PIVOT QUESTION

"Can we keep the LLM for language generation but supply the facts from an external source?"

Keep what the LLM is good at — language. Stop demanding what it cannot have — every fact.

Answer: **Yes — RAG →**

THE RAG IDEA · KNOWLEDGE OUTSIDE THE PARAMETERS

What Is External Knowledge?

BLACKBOARD ITEM 2

Parametric knowledge vs external knowledge

Definition: external knowledge means information **stored outside the model parameters**.

EXAMPLES

- PDF Documents
- Databases
- Web Pages
- Policies
- Research Papers
- FAQs
- Product Manuals
- Knowledge Bases
- Internal Documents

CONCEPTUAL DISTINCTION

MODEL PARAMETERS

learned **internal / parametric** knowledge

fixed at training time · implicit · not searchable

EXTERNAL KNOWLEDGE

information **retrieved when needed**

editable any time · explicit · searchable

THE RAG IDEA · DEFINITION

What Is RAG?

R etrieval - **A** ugmented - **G** eneration

A simple definition: RAG combines information retrieval with a generative LLM so that relevant external information can be supplied to the model while answering a query.

1 Retrieval

Find **relevant information**.

2 Augmented

Add that information to the LLM's **context**.

3 Generation

Use the LLM to produce the **final response**.

```
RAG = Retrieve + Augment + Generate
```

BASIC ARCHITECTURE · THE MAP OF EVERYTHING THAT FOLLOWS

RAG in One Diagram

1 **USER QUESTION**
student asks about the examination deadline

↓

2 **RETRIEVER**
searches for what matches the question

↓

3 **KNOWLEDGE SOURCE**
circulars, policies, manuals, records

↓

4 **RELEVANT INFORMATION**
the passage that actually answers the question

↓

5 **QUESTION + RETRIEVED CONTEXT**
the augmented prompt

↓

6 **LLM**
generates with the evidence in view

↓

GROUNDED RESPONSE
an answer tied to supplied evidence

MEMORY SENTENCE — HAVE STUDENTS REPEAT IT

**"Retrieve first.
Generate second."**

Read the diagram top to bottom:

- retrieval happens **before** generation, at query time
- the knowledge source sits **outside** the model
- the LLM sees the question **and** the evidence together
- the response is **grounded** in supplied information

This is the reference diagram for the whole module — Lecture 2 opens every box in this flow.

RUNNING EXAMPLE · THE EXAMINATION-FORM DEADLINE

The Same Question, With and Without RAG

Standalone LLM

QUESTION → LLM → ANSWER

"What is the last date for the **2026 Semester VIII examination form**?"

Might say: "Please check your university examination portal."

Or worse: "The deadline is 10 October." — without evidence

`Question` → `LLM` → `Answer`

Problem: no authoritative document supplied — **confidence should not be confused with correctness**.

With RAG

RETRIEVE → AUGMENT → GENERATE

Knowledge base contains `Examination_Circular_Oct_2026.pdf` with the passage:

"The last date for submission of Semester VIII examination forms is 18 October 2026."

`Student Question` → `Retriever` → `Find Relevant Circular Passage`

`Add Passage to Prompt` → `LLM` → `Answer`

Possible answer: **"The last date is 18 October 2026, according to the supplied examination circular."**

Much more **grounded** — the claim carries its evidence with it.

Main difference: RAG gives the model **additional evidence at inference time** — the answer depends on retrieved information plus prompt context, not on parameters alone.

BASIC ARCHITECTURE · WHO DOES WHAT

The Three Main Components

COMPONENT 1

Knowledge Source

Contains **information**.

PDFs | FAQs | Policies | Web pages | Database records

COMPONENT 2

Retriever

Finds the **most relevant information** for the user's question.

COMPONENT 3

Generator

Usually an **LLM** that uses **Question + Retrieved Context** to generate the answer.

BASIC MENTAL MODEL: Knowledge → Retriever → Context → Generator

BASIC ARCHITECTURE · SIX STEPS, ONE KEY WORD

The Basic RAG Workflow — and What "Augment" Means

THE SIX HIGH-LEVEL STEPS

1. User asks a question
2. Search external knowledge
3. Retrieve relevant information
4. **Add information to the prompt** — the augment step
5. Send augmented prompt to LLM
6. Generate response

"Next lecture we will **open every one of these boxes**."

ORIGINAL PROMPT

What is the examination deadline?

AFTER RETRIEVAL — THE AUGMENTED PROMPT

CONTEXT:

The examination circular states that Semester VIII forms must be submitted **by 18 October 2026**.

QUESTION:

What is the examination deadline?

This augmented prompt is then sent to the LLM.

Important: RAG usually provides information through the model's **context** — not by permanently writing that fact into the model **weights**.

BASIC ARCHITECTURE · PREVIEW & A COMMON MISUNDERSTANDING

Two Sides of a RAG System — Weights Stay Unchanged

A PREVIEW — LECTURE 2 TEACHES THE DETAILS

KNOWLEDGE PREPARATION SIDE

Documents → Prepare / Index → Searchable Knowledge Store

QUESTION-ANSWERING SIDE

Question → Retrieve → Context → LLM → Answer

Chunking, embeddings, vector databases, similarity search? **"Those are exactly what Lecture 2 is for."**

DOES RAG CHANGE MODEL WEIGHTS?

Model weights: **remain unchanged**

Instead — the **external context changes from query to query**:

MONDAY
Retrieve the **Monday circular**

FRIDAY
Retrieve the **updated Friday circular**

Same LLM. Different external information.

Takeaway: RAG enables **dynamic knowledge** without necessarily retraining the LLM.

COMPARISON 1 / 3 · WHAT ACTUALLY CHANGES

RAG vs Prompt Engineering

Prompt Engineering

Question + Better Instructions → LLM

EXAMPLE

"Answer in three bullet points."

RAG

Question + Retrieved Evidence → LLM

EXAMPLE

"Answer using this latest university circular."

Prompt Engineering → changes instructions | RAG → supplies information

COMPARISON 2 / 3 · CONNECTING MODULE 1 TO MODULE 2

RAG vs Fine-Tuning

RAG	Fine-Tuning
Supplies **external context**	Updates **trainable parameters**
Useful for changing **knowledge**	Useful for learned **behavior**
Knowledge can be **updated easily**	Requires **training** to change behavior
Retrieved **per request**	Behavior **persists** in model / adapters

Blackboard item 4: Prompt → **instructions** · RAG → **knowledge / context** · Fine-Tuning → **learned behavior**

Current examination deadline

→ **RAG**

University-specific complaint classification

→ **Fine-Tuning**

Both together

→ **RAG + Fine-Tuning**

COMPARISON 3 / 3 · RETRIEVAL ALONE IS NOT RAG

RAG vs Search Engine

Search

Question → Documents / Links

User reads the results.

RAG

Question → Retrieve Relevant Evidence → LLM Synthesizes Answer → Natural-Language Response

The machine reads, and answers for the user.

Important: RAG includes retrieval, but — **retrieval alone is not RAG**. Generation from retrieved context is the second part.

JUDGMENT · RAG IS NOT THE ANSWER TO EVERY PROBLEM

When RAG Is a Good Choice — and When It Is Not

RAG is attractive when knowledge is...

- frequently changing
- organization-specific
- private / proprietary
- too large to put manually in every prompt
- distributed across many documents
- expected to be source-grounded

EXAMPLES

current policies · internal manuals · documentation assistants · research libraries · customer-support knowledge · regulatory information

RAG may not be necessary when...

CASE 1 *"Answer politely."*

Use: **Prompting**

CASE 2 *"Always classify queries into our fixed internal schema."*

Likely: **Fine-Tuning / structured prompting** — depending on performance

CASE 3 Simple knowledge the model already handles reliably — **no grounding requirement**

RAG may add **unnecessary complexity**

Key point: use RAG when **external retrieval solves a real information problem** — not as a reflex.

BENEFITS · WHAT WELL-DESIGNED RAG CAN OFFER

Six Benefits of RAG

1

Fresher knowledge

Update documents rather than retraining everything.

2

Domain-specific information

Connect private knowledge sources.

3

Better grounding

Provide evidence relevant to the query.

4

Traceability

Possible to return sources and citations.

5

Lower knowledge-update cost

Change the knowledge base instead of modifying model weights.

6

Modularity

Retriever, knowledge source, and LLM can evolve independently.

✓ **Say:** RAG can **improve accuracy and grounding** when relevant, reliable evidence is retrieved.

✕ **Not:** "RAG always makes answers correct."

STUDENTS MUST HEAR THIS IN LECTURE 1

Limitations of RAG — and a Simple Failure Example

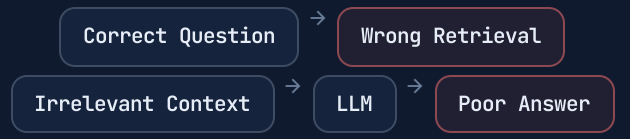
WHAT CAN GO WRONG

- **Wrong retrieval** → wrong context
- **Poor source documents** → poor evidence
- **Too much irrelevant context** → confusion / cost
- **Retrieval adds latency** → slower response
- **LLM can still hallucinate** → despite correct context
- **Knowledge permissions** → security / privacy challenge

Strong statement: RAG moves part of the problem from *"What does the LLM know?"* to **"Can we retrieve the right evidence?"** — that sets up Lecture 3.

A SIMPLE FAILURE EXAMPLE

User asks: **"What is the examination registration deadline?"**
Retriever accidentally returns: `Library Fine Policy`



Ask: *"Is this primarily a generation problem?"* — **No. It is first a retrieval problem.**

A PROFESSIONAL ENGINEERING POINT + THE UNIVERSITY CASE

RAG Is a System, Not a Single Model — One Architecture, Many Domains

RAG SYSTEM



Blackboard item 5: Good RAG = **Good Retrieval + Good Generation**. Quality depends on the complete pipeline — not only on the LLM.

EXAMPLE · UNIVERSITY RAG ASSISTANT

Academic Regulations | Examination Circulars | Fee Rules | Admission Policies | Student Handbook

Student asks: ***"Can I appear for the examination if my attendance is 72%?"***



Optionally: Source — Academic Regulations 2026, Section X

SAME ARCHITECTURE, DIFFERENT DOMAINS

Healthcare
Question → Medical Guidelines → LLM

Finance
Question → Current Policy → LLM

Customer Support
Question → Product Manual → LLM

Education
Question → Academic Regulations → LLM

The RAG architecture remains **similar**; the knowledge source and application requirements **change**.

CLASSROOM ACTIVITY · COMMIT BEFORE REVEALING

Prompt, RAG, Fine-Tuning, or Combination?

A	*"Answer every query in exactly three bullets."*	**Prompt**
B	*"Use today's updated examination circular."*	**RAG**
C	*"Always classify complaints into 15 fixed categories."*	**Fine-Tuning** may be appropriate if prompting alone is insufficient
D	*"Use today's circular and always output our fixed university schema."*	**RAG + Fine-Tuning**
E	*"Explain Python recursion."*	**Probably no RAG** needed — unless a specific external source is required

For each scenario, have students vote **before** revealing the verdict column — the disagreements are where the learning happens.

RUN AS TRUE / FALSE · HANDS UP FOR EACH

Seven Common Misconceptions

ANSWER KEY
F · T · F · F · T · F · T

1	RAG retrains the LLM for every question.	FALSE
2	RAG can provide information not represented in the base model.	TRUE
3	RAG completely eliminates hallucination.	FALSE
4	Retrieval alone is the complete RAG pipeline.	FALSE
5	A knowledge base can contain organization-specific documents.	TRUE
6	Fine-tuning and RAG solve exactly the same problem.	FALSE
7	Poor retrieval can lead to poor generation.	TRUE

Each false statement maps to a slide from today — ask students to **name the slide that corrects it**.

VOCABULARY TO KEEP · THE LECTURE IN ONE PICTURE

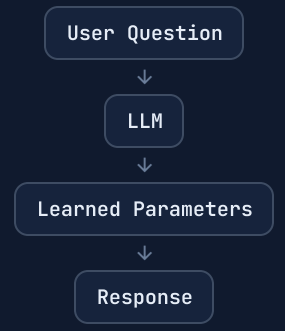
Six Terms to Remember — and the Complete Mental Model

SIX TERMS STUDENTS MUST REMEMBER

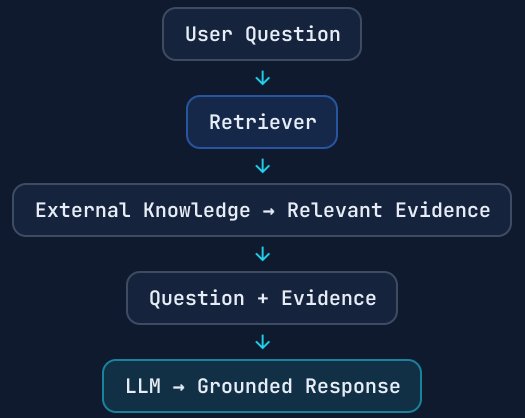
1. **Knowledge Base** — external information source.
2. **Retrieval** — finding relevant information.
3. **Retriever** — component that performs retrieval.
4. **Context** — retrieved information supplied to the LLM.
5. **Augmentation** — adding retrieved context to the model input.
6. **Generation** — producing the final answer.

Retrieve → Augment → Generate — **that is RAG.**

WITHOUT RAG



WITH RAG



WHAT STUDENTS SHOULD REMEMBER

Lecture Summary — Why RAG?

LLM KNOWLEDGE CAN BE…

- stale
- incomplete
- missing private information
- difficult to trace

THEREFORE

External Knowledge + Retrieval + LLM Generation → RAG

THE DEFINITION

RAG provides **relevant external information** to an LLM **at inference time.**

THE ONE SENTENCE TO LEAVE WITH

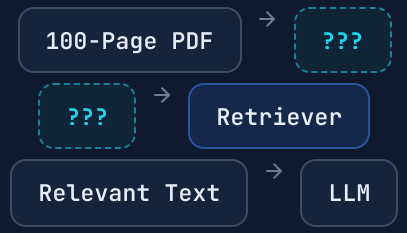
"Retrieval-Augmented Generation retrieves relevant information from an external knowledge source, adds that information to the LLM's context, and then uses the LLM to generate a more grounded response."

RAG = Retrieve + Augment + Generate

END WITH A PROBLEM, NOT A CONCLUSION

Bridge to Lecture 2 — Complete RAG Pipeline

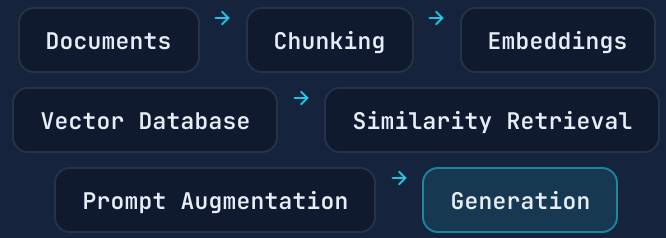
"We now understand the concept. But I have **deliberately hidden several boxes**."



Ask: *"How does a 100-page PDF become **searchable by meaning**?"*

NEXT

Lecture 2 — Complete RAG Pipeline



> "Today we learned **why** RAG exists. Next, we open the machine and understand exactly **how retrieval works**."